

N100 Financial Intelligence Platform

Analyst Guide

Fundamental analytics for the Nifty 100 constituents
92 companies | 30+ KPIs | 6 preset screeners | 16 API endpoints

Version 1.0 • 2026-07-28
Prepared by Raj Sarania, Data Analyst

1. What this platform is

A read-only analytics layer over the financial statements of the Nifty 100 companies present in the supplied datasets.

The platform takes twelve Excel datasets, loads them into a SQLite database, computes a standard set of financial KPIs, and exposes them four ways: an interactive dashboard, Excel workbooks, PDF tearsheets and a REST API. Nothing in it places trades or gives advice; it organises published financials so an analyst can compare companies quickly.

What is in the database

Table	Rows	What it holds
companies	92	Company master: name, sector links, about text
profitandloss	1,276	Revenue, operating profit, net profit, EPS by year
balancesheet	1,312	Equity, reserves, borrowings, assets by year
cashflow	1,187	Operating, investing and financing cash flow
financial_ratios	1,184	Computed KPIs: margins, returns, leverage, CAGRs
market_cap	552	P/E, P/B, EV/EBITDA, dividend yield (SIMULATED)
sectors	92	Broad sector, sub-sector, market cap category
peer_groups	56	11 peer groups with a benchmark company each
peer_percentiles	550	Percentile rank per metric within a peer group
documents	1,585	Annual report links filed with BSE

Read this before quoting any valuation number

The market_cap and stock_prices datasets are SIMULATED. Every P/E, P/B, EV/EBITDA, dividend yield and market cap figure in this platform derives from them. They demonstrate the calculations working; they are not market data. Anything labelled SIMULATED must not be used to justify an investment decision.

Getting set up

```
python -m venv .env
.venv\Scripts\activate # Windows
source .venv/bin/activate # macOS or Linux
pip install -r requirements.txt
```

The repository already contains a populated database and every generated output, so you can go straight to the dashboard. Rebuild from the source Excel files only if the data changes.

```
make load # Excel to SQLite
make ratios # populate financial_ratios
make screener # screener, peer ranks, radar charts
make valuation # valuation summary and flags
make nlp # parse analysis text, generate pros/cons
make cashflow # cash flow intelligence
make tearsheets # 91 company PDFs, 11 sector PDFs, portfolio
make test # full test suite
```

If make is not installed, run any target directly, for example `python -m src.screener.run_screener`.

2. Using the screener

The fastest way to narrow 92 companies down to a shortlist.

Start the dashboard and open the Screener screen from the sidebar:

```
streamlit run src/dashboard/app.py
# then open http://localhost:8501
```

The two ways to screen

- **Preset buttons** fill in the sliders for a named investment style. Click Quality, Value, Growth, Dividend, Debt-Free or Turnaround.
- **Sliders** let you set any of ten thresholds yourself. The results table updates as you move them.

A slider parked at its neutral end is not applied, so leaving a metric alone does not silently exclude companies that happen to be missing that figure.

The six presets

Preset	Looks for	Returns
Quality Compounder	ROE above 15%, D/E under 1, positive free cash flow, revenue growth above 10%	22 companies
Value Pick	P/E under 20, P/B under 3, dividend yield above 1%	2 companies
Growth Accelerator	PAT CAGR above 20%, revenue CAGR above 15%	18 companies
Dividend Champion	Dividend yield above 2%, payout under 80%, positive free cash flow	30 companies
Debt-Free Blue Chip	Zero borrowings, ROE above 12%, revenue above 5,000 Cr	2 companies
Turnaround Watch	Revenue CAGR 3yr above 10%, positive free cash flow, falling leverage	34 companies

Why Value Pick and Debt-Free return so few companies

Both are working correctly. Value Pick screens on P/E and P/B, which are uncorrelated in the simulated dataset ($r = -0.11$) where in real markets they move together: 15 companies pass the P/E test and 9 pass the P/B test, but only 2 pass both. Debt-Free requires literally zero borrowings, and only 3 of 92 companies report that. Relaxing D/E to below 0.05, the conventional market definition of debt free, returns 16. Both calibrations are documented in `config/screener_config.yaml`.

Two rules that are not on the sliders

- **Banks are exempt from the D/E ceiling.** A debt-to-equity limit does not reject Financials, because leverage of 8 is structurally normal for a lender. In the Excel export those cells appear red: the company was admitted by the rule, not by the number.
- **No interest expense means infinite coverage.** A company with no debt has no interest coverage ratio to compute. It passes any coverage minimum rather than failing on a blank.

Exporting

The Download CSV button exports exactly the rows and columns on screen. For the full preset run with colour coding, use `output/screener_output.xlsx`, which has one sheet per preset plus a summary.

The dashboard presets are an approximation

Three preset rules cannot be expressed as a slider: an equality test (D/E exactly zero), a payout cap, and a year-on-year direction. The dashboard therefore returns more companies than the workbook for Debt-Free (33 vs 2), Dividend (33 vs 30) and Turnaround (40 vs 34). Run make screener for the authoritative result.

3. The eight dashboard screens

What each screen is for and what to watch out for on it.

1. Home

Index-level summary: six KPI tiles, a sector donut, the top five companies by composite quality score, and a table of sector medians. The sidebar year selector re-points the valuation tiles between 2019 and 2024.

Average ROE reads 122%

That is not a bug in the calculation. A handful of companies report equity figures that are not on the same scale as their profit, which inflates the mean. The tile tooltip gives the median, 15.8%, which is the figure to quote. All such cases are itemised in `output/ratio_edge_cases.log`.

2. Company Profile

One company in detail. Search by name or ticker, then read the company card, six KPI tiles, a ten-year revenue and profit bar chart, a dual-axis ROE and ROCE line chart, and the generated pros and cons. A company with fewer than ten years of data shows what it has, with a note saying so.

3. Screener

Covered in chapter 2.

4. Peer Comparison

Pick one of the 11 peer groups and see any member drawn against the peer average on eight axes, plus a side-by-side table with the benchmark company highlighted in amber. The radar plots percentile ranks, not raw values, because an ROE of 30 and a D/E of 0.3 cannot share a radius. Debt to equity is inverted, so further from the centre is better on every axis.

5. Trend Analysis

Overlay up to three metrics across ten years. Each point is annotated with its change against the prior year. Each metric gets its own axis so a percentage and a crore figure can share the chart.

6. Sector Analysis

Revenue against ROE as a bubble chart, sized by market cap and coloured by sub-sector, with sector median bars underneath. Companies without a market cap are drawn at the smallest bubble size rather than dropped.

7. Capital Allocation

A treemap of every company grouped by what it does with its cash, derived from the signs of operating, investing and financing cash flow. Click a pattern to list its members.

Pattern	CFO	CFI	CFF	Reading
Reinvestor	+	-	-	Generates cash, reinvests it
Shareholder Returns	+	-	-	As above but CFO exceeds profit
Cash Accumulator	+	+	+	Cash in from everywhere
Liquidating Assets	+	+	-	Selling assets, repaying capital
Growth Funded by Debt	-	-	+	Investing while operations consume cash
Mixed	+	-	+	Generates cash, invests, still raises funds
Distress Signal	-	+	+	Burning cash, selling assets, borrowing
Pre-Revenue	-	-	-	Cash out across the board

Two companies fall into a ninth sign combination that the eight standard patterns do not name, and show as Unknown Pattern.

8. Annual Reports

Available report years with clickable BSE PDF links. 52 links are missing from the source data and show a red Report unavailable badge. Link checking against bseindia.com is opt-in, because verifying every URL on each page load would take the screen past its three second budget.

4. Reading a tearsheet

The two-page PDF summary produced for each company.

Tearsheets live in `reports/tearsheets/`, one per company, named `{TICKER}_tearsheet.pdf`.

Page 1

- Navy header with the company name, ticker and sector
- Six KPI tiles: ROE, ROCE, net profit margin, D/E, revenue CAGR and free cash flow
- Ten-year revenue and net profit bars
- ROE and ROCE on a dual axis, so a scale difference between them does not flatten one line

Page 2

- Balance sheet composition: net worth, borrowings and other liabilities stacked by year
- Cash flow waterfall for the latest year, ending in net cash flow
- A capital allocation badge naming the pattern
- Pros in green and cons in red, each with a confidence percentage

What the confidence percentage means

Each pro and con comes from a rule. The percentage is how strongly the company clears that rule's threshold, not a probability of anything happening. A rule reports 60 when a company only just qualifies and 100 when it clears the bar comfortably. Only statements scoring above 60 are printed.

Two rule ids to treat differently

PRO-13 and CON-13 are relative, not absolute. They appear only when a company triggered no absolute rule at all, and they name its strongest or weakest metric against its sector. A CON-13 reading "no absolute red flag, but ranks in only the 40th percentile" is not a warning; it means nothing was found. 47 of the 92 companies trip no absolute con rule because they are genuinely clean.

Generating tearsheets

```
# every company, plus sector and portfolio PDFs
make tearsheets

# a specific few
python -m src.reports.tearsheet TCS HDFCBANK RELIANCE
```

The batch takes a few minutes because each tearsheet renders four charts. Companies with fewer than three years of data are skipped and logged to `output/skipped_tearsheets.csv`; at present that is one company, JIOFIN.

The other PDF outputs

Output	Location	Contents
Sector reports	<code>reports/sector/</code>	One PDF per sector: median KPIs and every company with 8 metrics
Portfolio summary	<code>reports/portfolio/</code>	One page per company with six KPIs and trend arrows
Radar charts	<code>reports/radar_charts/</code>	One PNG per company against its peer or index average

In the portfolio summary, a green triangle means the metric improved against the prior year, red means it declined, and a grey marker means the change was within 2% either way. For debt to equity a fall counts as an improvement.

5. Calling the API

16 read-only endpoints under /api/v1.

```
uvicorn src.api.main:app --port 8000
# interactive docs at http://localhost:8000/docs
```

The API and the dashboard can run at the same time; they use different ports and there is no conflict.

Endpoints

Endpoint	Returns
GET /health	Status, row counts for all 10 tables, uptime, version
GET /companies	All 92 companies; filter by sector, market_cap_category or search
GET /companies/{ticker}	Full profile: master record, sector, latest KPIs
GET /companies/{ticker}/pl	Profit and loss history
GET /companies/{ticker}/bs	Balance sheet history
GET /companies/{ticker}/cashflow	Cash flow history
GET /companies/{ticker}/ratios	Computed KPIs per year
GET /companies/{ticker}/tearsheet	The tearsheet PDF as a download
GET /companies/{ticker}/peers/compare	Radar data: 8 axes, peer average, benchmark
GET /companies/{ticker}/documents	Annual report links with a validity flag
GET /screener	Ranked results for any threshold combination
GET /sectors	Every sector with counts and medians
GET /sectors/{sector}/companies	Companies in a sector with latest KPIs
GET /peers/{group_name}	Peer group with a percentile rank per metric
GET /market-cap/{ticker}	Valuation multiples 2019 to 2024 (SIMULATED)
GET /portfolio/stats	P10 to P90, mean and std for 10 KPIs

Worked examples

```

# is the service healthy?
curl http://localhost:8000/api/v1/health

# quality companies: ROE above 15, D/E under 1
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1"

# run a named preset instead of raw thresholds
curl "http://localhost:8000/api/v1/screener?preset=quality_compounder"

# one company in full
curl http://localhost:8000/api/v1/companies/TCS

# ten years of computed ratios
curl http://localhost:8000/api/v1/companies/TCS/ratios

# a single year
curl "http://localhost:8000/api/v1/companies/TCS/ratios?year=Mar%202024"

# profit and loss for a date range
curl "http://localhost:8000/api/v1/companies/TCS/pl?from_year=2020-03&to_year=2024-03"

# download a tearsheet
curl -o tcs.pdf http://localhost:8000/api/v1/companies/TCS/tearsheet

# peer group with percentile ranks
curl "http://localhost:8000/api/v1/peers/IT%20Services"

# index-wide percentile table
curl http://localhost:8000/api/v1/portfolio/stats

```

Status codes

Code	When
200	Success
400	A threshold is out of range, a preset name is unknown, or a year is not YYYY-MM
404	Unknown ticker, sector or peer group; or no tearsheet exists for that company
422	FastAPI rejected the query type, for example limit=0

A company with no peer group returns HTTP 200 with the message "No peer group assigned" rather than a 404. 37 of the 92 companies are in that position, and it is a normal answer rather than an error.

The OpenAPI 3 specification is exported to docs/openapi.json and can be imported into Postman.

6. Company archetypes and statistics

What the clustering says and how to read the supporting files.

Every company is assigned to one of five archetypes by KMeans clustering on five features: ROE, debt to equity, revenue CAGR, free cash flow CAGR and operating margin. Assignments are in `output/cluster_labels.csv`.

Archetype	Companies	Profile
High-Quality Compounders	10	Very high returns with low leverage
High-Margin Franchises	13	Operating margins near 80%, moderate returns
Emerging Growth	26	Strong revenue and cash flow growth
Defensive Dividend Payers	28	Mature, slower growth, low leverage
Leveraged Financials	15	Banks and NBFCs, mean D/E near 7

Why two names differ from the project document

The document offers five example archetype names. Two do not describe what the clustering actually found: the most leveraged cluster is entirely banks and NBFCs, so it is named for its leverage rather than called Emerging Growth; and one cluster carries 80% margins with modest returns, which makes it a high-margin group rather than a distressed one. No cluster in this universe is genuinely distressed, since the weakest still averages 13% ROE.

Supporting files

File	What it tells you
<code>output/portfolio_stats.csv</code>	P10 to P90, mean and standard deviation for 10 KPIs. Use it to judge whether a company is unusual before quoting a number
<code>output/outlier_report.csv</code>	Companies more than 3 standard deviations from their sector mean on any metric
<code>output/cluster_profiles.csv</code>	Mean and median of each feature per cluster
<code>reports/elbow_plot.png</code>	Inertia against k, showing why five clusters was chosen
<code>reports/correlation_heatmap.png</code>	Pearson correlation of 10 KPIs across the index
<code>output/cashflow_intelligence.xlsx</code>	CFO quality, CapEx intensity, distress and deleveraging flags per company
<code>output/distress_alerts.csv</code>	Companies burning operating cash while raising financing

The distress flag over-reports lenders

The flag fires when operating cash flow is negative and financing cash flow is positive. For a bank or NBFC that is the ordinary signature of loan book growth, not trouble: 9 of the 13 flagged companies are profitable Financials, including one with 14,451 Cr of net profit. `distress_alerts.csv` carries a `structurally_normal_for_sector` column that separates those from the four worth reviewing.

Read the correlation heatmap with care

Pearson correlation is sensitive to the extreme source values documented in `ratio_edge_cases.log`. ROE against asset turnover reads +0.96 on Pearson but only +0.49 on rank correlation, and net profit margin against interest coverage reads +0.79 against +0.09. Where those disagree, the coefficient reflects a few companies rather than a relationship across the index. The figure carries a footnote naming the worst offenders.

7. Troubleshooting

Things that look wrong but are not, and things that are.

Numbers that look wrong but are correct

What you see	Why	What to do
A ratio shows N/A	The input was missing or the denominator was zero. The engine returns nothing rather than dividing by zero	Check the company on the Profile screen; the note says how many years exist
Interest coverage is blank	The company has no interest expense, so there is nothing to divide by. It is debt free	Read it as infinite coverage. It passes coverage filters
PAT CAGR is blank but revenue CAGR is not	The company was loss-making in the base year, so a growth rate would be meaningless	Nothing. This is the documented TURNAROUND case
Average ROE is 122%	A few companies report equity on a different scale from profit	Use the median, 15.8%. See ratio_edge_cases.log
A bank passed a D/E filter it should have failed	Financials are exempt from D/E ceilings by design	Nothing. Red cells in the Excel export mark these rows
A profitable company is flagged distressed	It is a lender, and lending growth looks like negative operating cash flow	Check structurally_normal_for_sector in distress_alerts.csv

Things that are actually broken

Symptom	Cause	Fix
Dashboard says the database was not found	data/nifty100.db is missing	Run make load then make ratios
A screen errors on a missing column	The ratio engine has not been run since a schema change	Run make ratios
Capital Allocation screen is empty	output/capital_allocation.csv has not been generated	Run make ratios
Tearsheet endpoint returns 404	That company was skipped for having under 3 years of data, or the batch has not run	Check output/skipped_tearsheets.csv, then run make tearsheets
Clusters endpoint returns 0	cluster_labels.csv has not been generated	Run python -m src.analytics.clustering
Two tickers cannot be found in any search	ULTRACEMCO and UNIONBANK appear in the statements but have no company master record	A source data defect. They still appear in aggregates

Known data limitations

- **companies.xlsx is truncated.** It holds 92 rows and stops part way through the alphabet, so eight tickers appear in the financial statements with no master record.
- **SBIN has no computed financials** despite being the designated Public Sector Banks benchmark, so that peer group has no highlighted benchmark row.
- **The analysis table covers 5 companies**, not 92. It is used to cross-check the CAGR engine, not as a data source.
- **prosandcons covers 16 companies**, so all generated pros and cons come from financial rules rather than supplied text.
- **There are 10 broad sectors, not 11.** The eleventh entry in sector listings is the Unclassified bucket.
- **119 duplicate company-year rows** exist in the source statements. Analytics de-duplicate before counting.

- **Operating profit stands in for EBITDA**, because the datasets carry no depreciation line.

Where to look next

Question	Document
Why was a ratio implemented this way?	output/ratio_edge_cases.log
What was decided in a given sprint?	docs/Sprint2_retrospective.md through Sprint6
Is the platform meeting its acceptance criteria?	docs/acceptance_checklist.pdf
How fast is it, and where is the cost?	output/perf_notes.md
What does the test suite cover?	reports/pytest_report.html

One closing note

This platform organises published financial statements. It does not value companies, forecast anything, or recommend action. The rule confidence scores measure how strongly a threshold was cleared, not the likelihood of any outcome. Every valuation metric comes from simulated data. Treat the output as a starting point for analysis, not a conclusion.